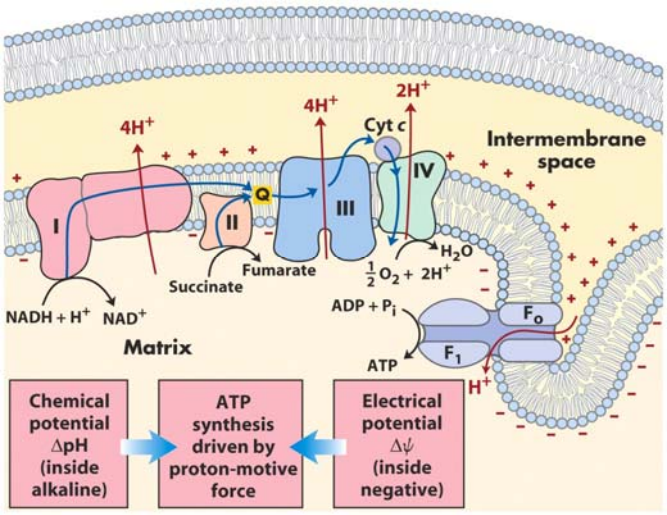
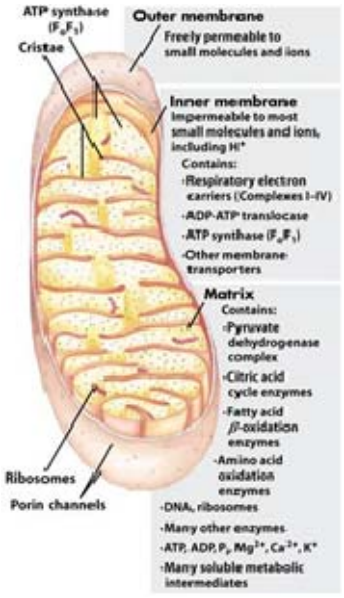
Oxidative Phosphorylation



- Humans use about **83 kg of ATP daily**
- However, **the available ATP deposits are approximately 250g**
- **This huge disparity** between the needs and the availability of ATP **is solved by recycling ADP to ATP**
- Each **ATP molecule is recycled approximately 300 times/day**, through a process called **oxidative phosphorylation**
- **Def.** Oxidative phosphorylation is the process by which **ATP is formed as a result of the transference of electrons, from NADH or $FADH_2$ to O_2, through electronic transporters located in the inner mitochondrial membrane.**
- **The electrons flux gives rise to the pumping of H^+ from the mitochondrial matrix to outside,** which generates a H^+ gradient**. The inner mitochondrial membrane is impermeable to H^+.**
- ATP is synthesized when protons return to the mitochondrial matrix through an enzyme complex called ATP synthase
- Oxidative phosphorylation is the culmination of cellular respiration
- The skeletal carbons of organic fuels are oxidized to release electrons with a high transference potential (NADH and $FADH_2$), in other words, they constitute an electron motive force
- This electron motive force is converted into a proton motive force by the respiratory chain complexes that pumps H^+ out of the mitochondria.
- Finally, the proton motive force drives ATP synthesis

Mitochondria: the scenario of the action



- Mitochondria have two membranes
- **The outer membrane is permeable** to small molecules and ions that move from one side to the other through channels called porins
- **The inner membrane is impermeable** even to ions such as H^+.
- The traffic of molecules to one side or the other is done through specific selective channels.
- **The inner membrane contains the components of the electron transport chain and ATP synthase**
- **The mitochondrial matrix contains both the pyruvate dehydrogenase complex and the enzymes of Krebs cycle, β-oxidation (oxidation of fatty acids) and the oxidation of amino acids.**

Electron funneling to the respiratory chain

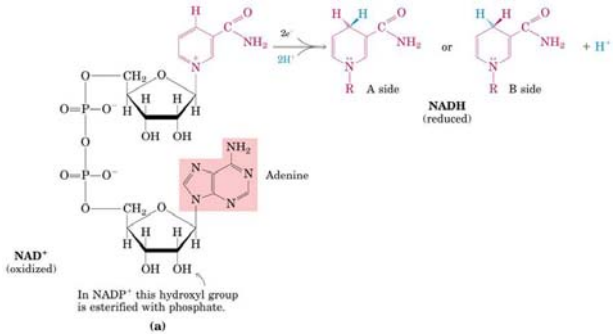
- Most of the electrons that drive ATP synthesis come from the action of dehydrogenases that oxidize substrates by catabolic reactions and channel them to common electronic transporters: NADH and $FADH_2$

TABLE 19-1 Some Important Reactions Catalyzed by NAD(P)H-Linked Dehydrogenases

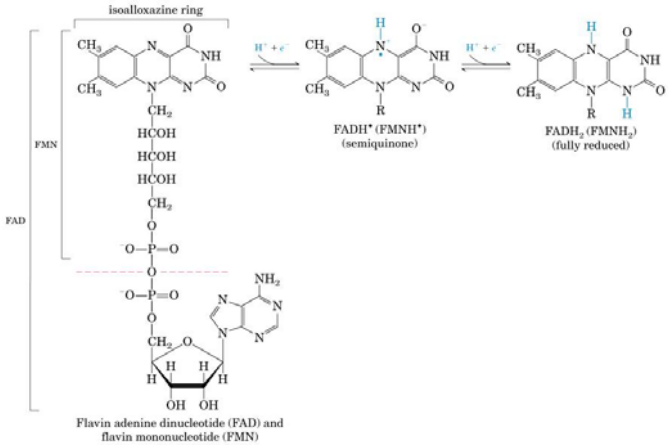
Reaction*	Location†
NAD-linked	
α-Ketoglutarate + CoA + NAD^+ ⟶ succinyl-CoA + CO_2 + NADH + H^+	M
L-Malate + NAD^+ ⟶ oxaloacetate + NADH + H^+	M and C
Pyruvate + CoA + NAD^+ ⟶ acetyl-CoA + CO_2 + NADH + H^+	M
Glyceraldehyde 3-phosphate + P_i + NAD^+ ⟶ 1,3-bisphosphoglycerate + NADH + H^+	C
Lactate + NAD^+ ⟶ pyruvate + NADH + H^+	C
β-Hydroxyacyl-CoA + NAD^+ ⟶ β-ketoacyl-CoA + NADH + H^+	M
NADP-linked	
Glucose 6-phosphate + $NADP^+$ ⟶ 6-phosphogluconate + NADPH + H^+	C
NAD- or NADP-linked	
L-Glutamate + H_2O + $NAD(P)^+$ ⟶ α-ketoglutarate + NH_4^+ + NAD(P)H	M
Isocitrate + $NAD(P)^+$ ⟶ α-ketoglutarate + CO_2 + NAD(P)H + H^+	M and C

*These reactions and their enzymes are discussed in Chapters 14 through 18.
†M designates mitochondria; C, cytosol.

- Most dehydrogenases use NAD^+ as the electron acceptor.
- These dehydrogenases remove two hydrogen atoms from their substrates. One is transferred as hydride ion ($:H^-$) and the other appears as H^+ in the medium



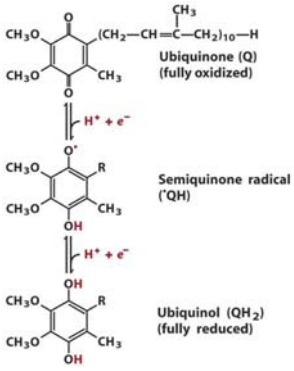
- The other group of dehydrogenases use FAD as electron acceptor. These types of enzymes are called flavoproteins.



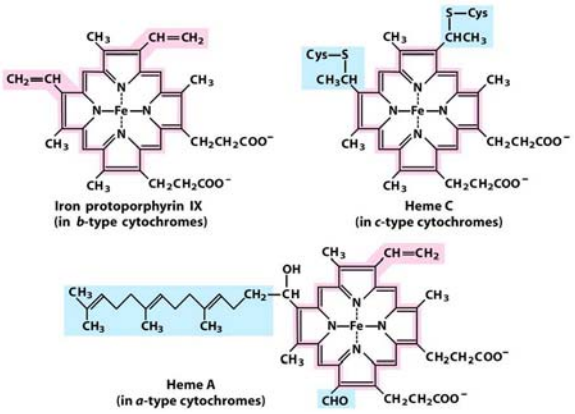
Respiratory electron carriers at the inner mitocondrial membrane

- The mitochondrial respiratory chain is made up of a series of electronic transporters that act sequentially
- Most of them are integral proteins with prosthetic groups capable of accepting and giving up one or two electrons
- In addition to NAD and flavoproteins there are three additional types of electron transport molecules:

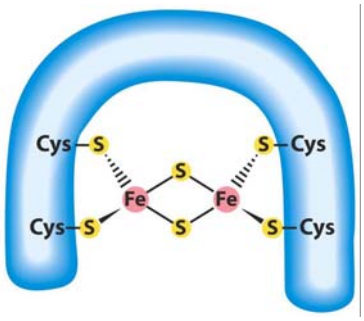
o **ubiquinone or coenzyme Q**: it can accept one electron (semiquinone radical * QH), or two electrons (ubiquinol QH2). As it is a small and hydrophobic molecule, it can diffuse freely within the lipid bilayer of the inner membrane and can transfer electrons from one complex to another.



o **cytochromes:** they are **proteins** with an **heme group (prostetic group)**. In mitochondria there are three classes of cytochromes designated as a, b, and c. They are strongly bound to proteins.
Cytochrome c is an exception with respect to other cytochromes since it is bound to a soluble protein which allows it to transfer electrons from one complex to another, such as CoQ

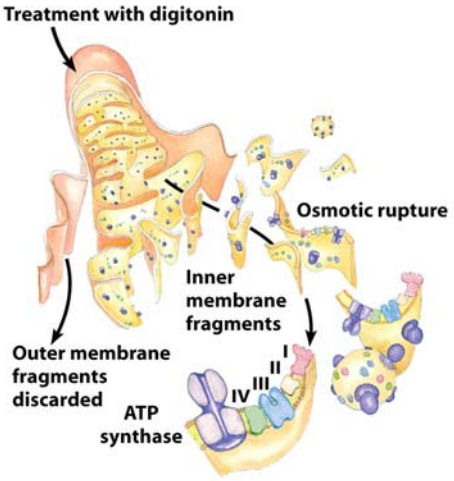


o **Iron-sulfur proteins**: iron is present in association with inorganic sulfur atoms or with Cys residues

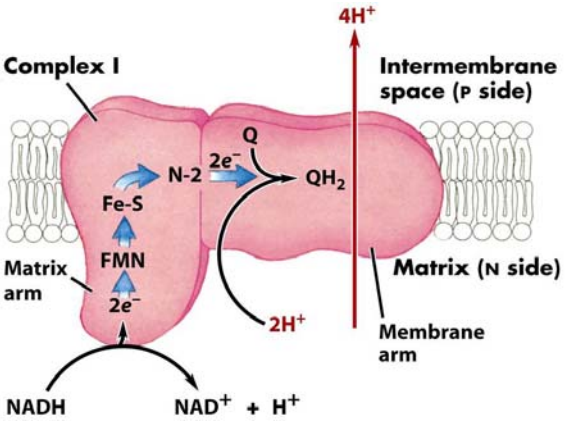


Arrangement of the electron carriers in multienzyme complexes

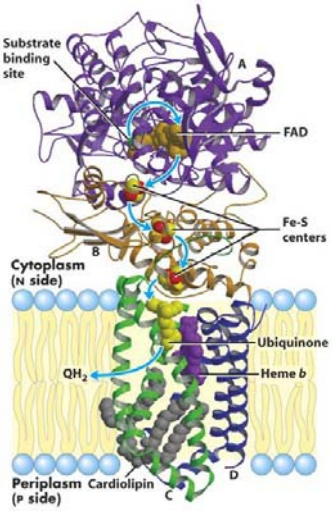
- The electron carriers of the respiratory chain are organized into membrane-embedded in the inner membrane.
- When the inner membrane is solubilized with mild detergents, four complexes are isolated



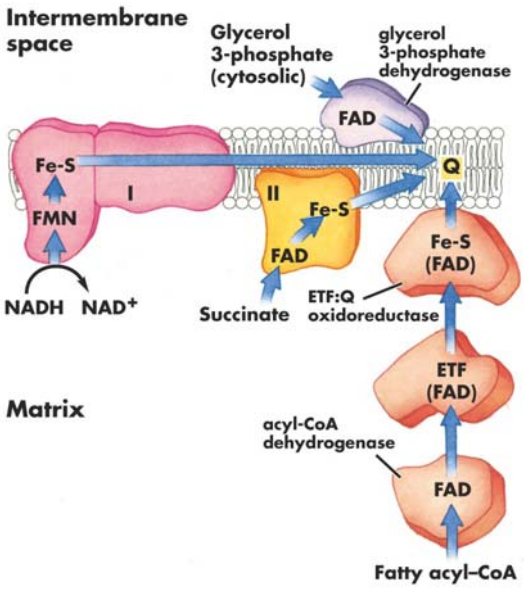
- **Complex I or NADH-ubiquinone oxidoreductase**: it is composed by 43 polypeptidic chains. As cofactors for electronic transport contains FMN and Fe-S centers
- It catalyzes two simultaneous coupled processes: the reduction of ubiquinone with electrons from NADH and the transfer of four protons from the matrix to the intermembrane space.
- In other words, complex I is a proton pump drived by the energy released during the electron transference
- This complex can be inhibited with amital (barbiturate), rotenone (insecticide) and piericidin A (antibiotic)
- **Subsequently, the reduced ubiquinone, due to its hydrophobic character, diffuses and translate the electrons to complex III**



- **Complex II or Succinate dehydrogenase**: it is the unique membrane-bound enzyme of the Krebs cycle. It contains a cytochrome b, several Fe-S and FAD centers.
- Complex II **couples the oxidation of succinate with the reduction of ubiquinone**.
- Electron transfer through complex II is not accompaniedby proton pumping across the inner membrane.
- In resume, electrons from succinate pass through complex II and finally reduce the coenzyme Q. Conenzyme Q transfers the electrons to complex III

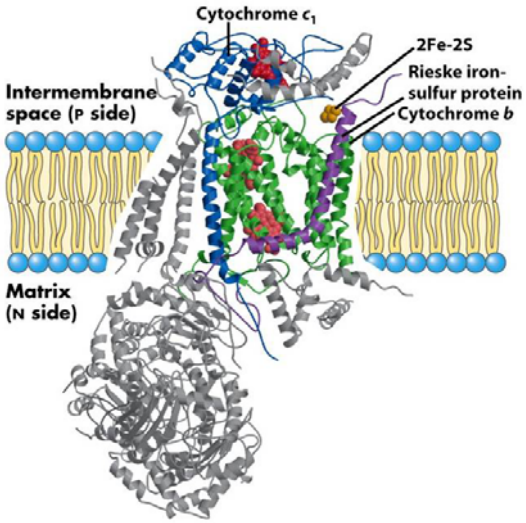


- In summary, electrons stripped from organic fuels are incorporated into the respiratory chain through complex I or through complex II, and from these to ubiquinone (Coenzyme Q)

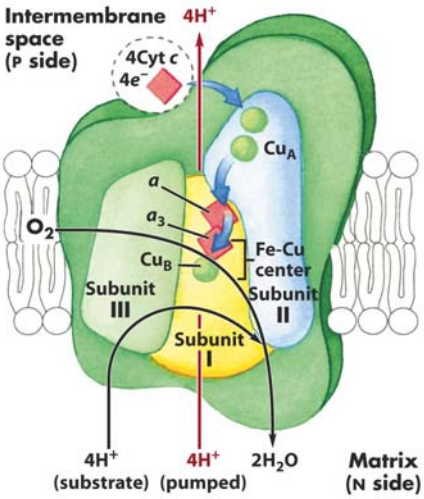


- **There are other alternative electron incorporations to the respiratory chain**:
- The electrons released at the first oxidation of the fatty acids β oxidation (mediated by acyl-CoA dehydrogenase) are transferred from $FADH_2$ to a protein called electron transfer flavoprotein (ETFP) and then to ubiquinone.

- Another way is glycerol 3-phosphate dehydrogenase (G3PDH), which oxidizes glycerol, located in the inner mitochondrial membrane. It also incorporates electrons by reducing ubiquinone. This enzymatic system of G3PDH is one of the processes that allows the transfer of electrons generated in the cytoplasm by glycolysis (NADH) to the mitochondrial respiratory chain

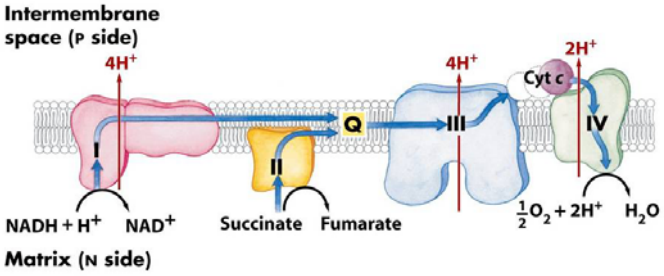


- **Complex III or ubiquinone-cytochrome c oxidoreductase**; couples the electrons transfer from ubiquinol to cytochrome c with the concomitant transport of protons from the matrix to the intermembrane space.
- **- The electrons finish their pathway by reducing cytochrome c and this transfers them to complex IV**



- **Complex IV or cytochrome oxidase**: transports electrons from cytochrome c to molecular oxygen.There is a simultaneous pumping of protons to the intermembrane space

Summary of the electronic transport process



Radical formation during electron transport

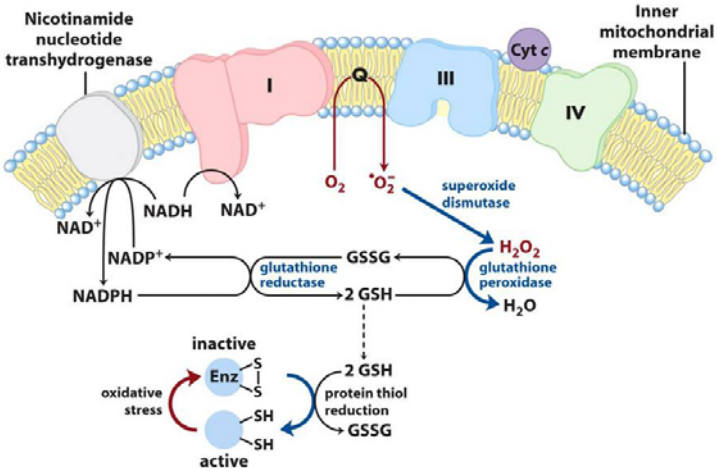
- Electronic transport is not without risks for the cell, since during it there is the possibility that **some electrons "escape" from the process** and end up partially reducing oxygen, generating ROS (reactive oxygen species)

$$O_2 + e^- \rightarrow {}^*O_2^-$$

- The superoxide radical is highly reactive and is capable of generating *OH radicals that are even more reactive and can damage enzymes, membrane lipids, and nucleic acids.
- To protect themselves from ROS the cells have some batteries of enzymes
- One of the protective enzymes is **superoxide dismutase** (SOD)

$$2\ {}^*O_2^- + 2H^+ \rightarrow H_2O_2 + O_2$$

- Hydrogen peroxide is then neutralized thanks to **glutathione peroxidase**

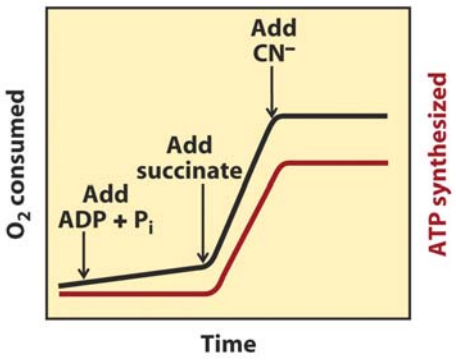


- **This defense mechanism is highly dependent on the availability of NADPH and glutathione (GSH).**
- Glutathione is a peptide formed **by Glu-Cys-Gly**. The SH groups of glutathione are those that reduce hydrogen peroxide to water, but in this reaction they are oxidized, forming a disulfide bridge with another glutathione molecule (GS-SG).
- **NADPH is necessary to reduce GS-SG to 2GSH** (active glutathione) **by glutathione reductase**

- **Furthermore, reduced glutathione is an essential factor in the maintenance of the –SH groups in proteins in a reduced state, which are necessary to keep their activities intact.**

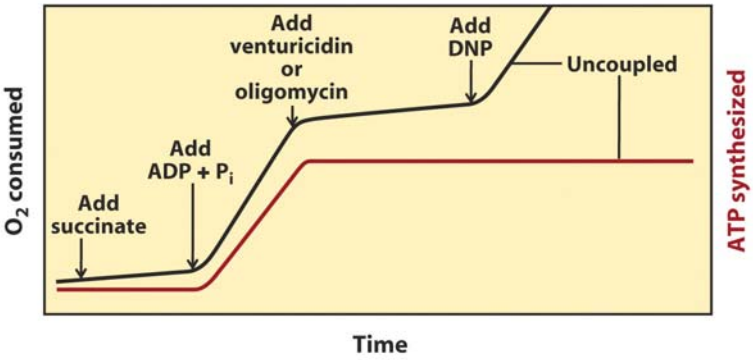
ATP synthesis

- The question we need to answer is, what chemical mechanism couples the flow of protons with the synthesis of ATP?
- The model that is accepted for this coupling **is the Mitchell's chemosmotic model**, according to which the electrochemical energy derived from the proton gradient, generated during electron transport, drives ATP synthesis as the protons passively flow into the matrix through an enzyme called ATP synthase located in the inner mitochondrial membrane



- **The coupling between electron transport and ATP synthesis can be demonstrated experimentally**
- When mitochondria are suspended in a buffer with ADP and Pi and an oxidizable substrate (succinate), three quantifiable processes take place:
  - **the substrate is oxidized: succinate → fumarate**
  - **O_2 is consumed**
  - **ATP is synthesized**

- In other words, the consumption of O_2 and the synthesis of ATP depends on the presence of an oxidizable substrate (electron donor)
- If, during the process, we add an electron transport inhibitor, such as CN^- (it inhibits complex IV), ATP synthesis is also blocked

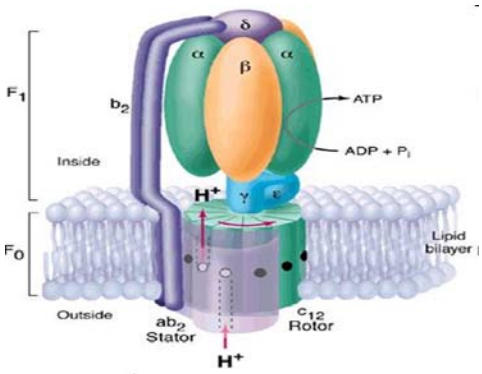


- The same happens if it is done the other way around; blocking ATP synthesis with ATP synthase inhibitors such as oligomycin or venturicidin, inhibits electron transport (oxygen is not consumed)

- To demonstrate that the proton gradient is responsible for the synthesis of ATP, a weak hydrophobic acid, such as dinitrophenol (DNP), can be added to the mitochondria.

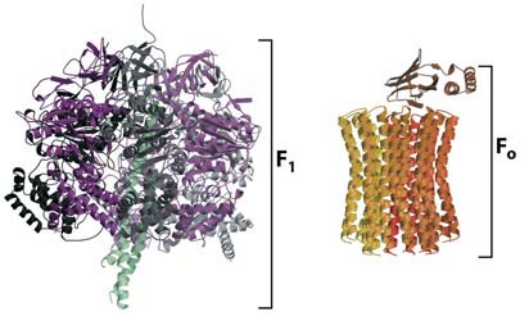
DNP will be charged with protons in the extramitochondrial space and, crossing the inner membrane, will release them into the matrix. This process uncouples the mitochondria, that is, it breaks the proton gradient generated by the transport of electrons, preventing ATP synthesis.

ATP synthase structure

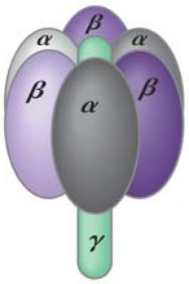


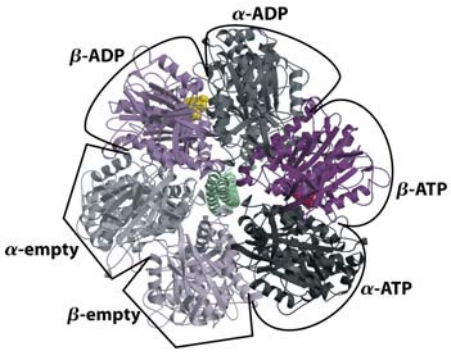
- ATP synthase, which is also called complex V, is a large enzymatic complex at the inner mitochondrial membrane that catalyzes the formation of ATP from ADP and Pi, accompanied by the flow of protons from the intermembrane space to the matrix
- It has two components: F1 and Fo

    - Fo is an integral membrane protein that allows protons to pass from one side of the inner membrane to the other
    - F1 is oriented towards the mitochondrial matrix and contains the catalytic centers.

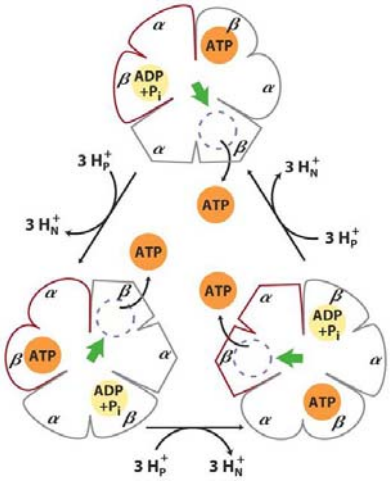


- F1 has nine subunits with a composition $\alpha_3\beta_3\gamma\delta\varepsilon$
- Each of the β subunits has a catalytic site for ATP synthesis
- The γ subunit acts as the central axis of F1, with a domain that interacts directly with any of the three β subunits.
- The interactions between the γ and β subunits cause that, although the amino acid sequence is the same in the three β subunits, the conformation is not the same
- Each of the three β subunits can be found in three different conformational states at the ADP / ATP binding sites.
    - **β empty**: open (low affinity, allows the release of ATP)
    - **β ADP**: light (intermediate affinity, allows the binding of ADP and Pi)
    - **β ATP**: tight (high affinity, ATP synthesis occurs)

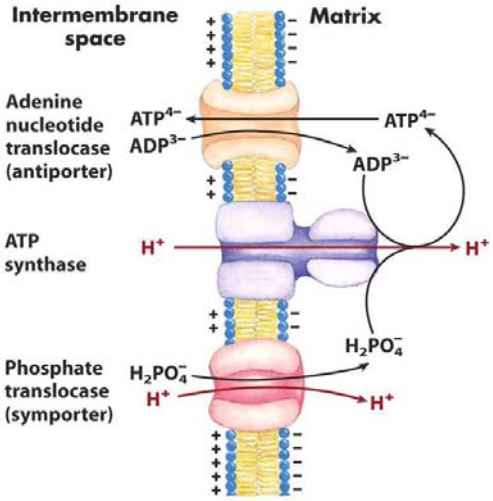




Catalitic mechanism for ATP synthase



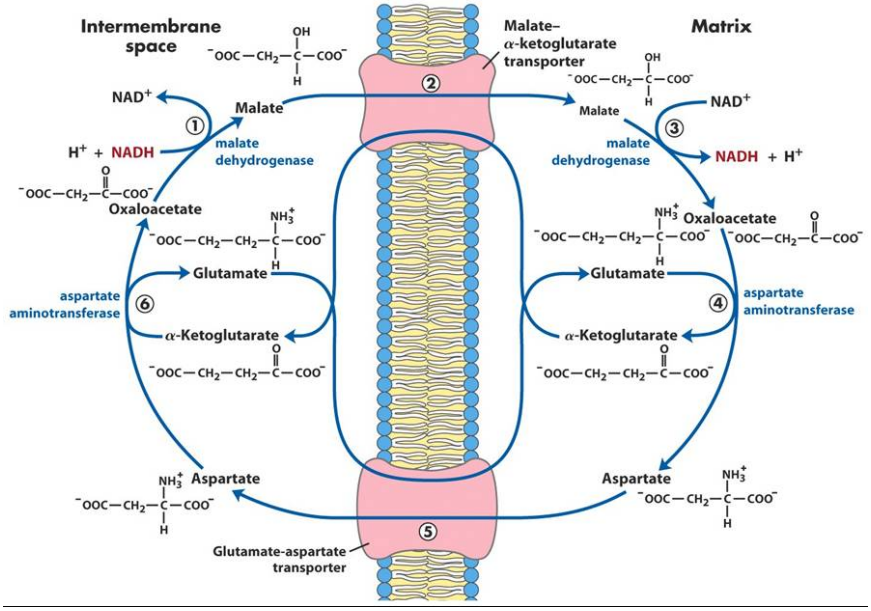
- The hypothesis about the catalytic mechanism of ATP synthase was proposed by Paul Boyer and is called **rotational catalysis**.
- The three active sites of F1 alternate in the catalysis of ATP synthesis
- The passage of protons through Fo causes the cylinder and the γ subunit attached to it to rotate.
- With each rotation of 120º, **the γ subunit comes into contact with a different β subunit, which causes a conformational change**.
- A determined β subunit begins the cycle in the β-ADP conformation, thus binding ADP and Pi
- Then, due to the passage of protons and the turn of γ, the subunit β-ADP changes its conformation, adopting the form β-ATP
-Finally, again due to the passage of protons and the turn of γ, the β-ATP subunit, changes towards the β-empty conformation



- Taking into account that ,the electronic transfer from NADH pumps10 H^+ to the intermembrane space, and the transfer from $FADH_2$ pumps $6H^+$, and that to synthesize 1 ATP 4 H^+ are necessary (one is used for the transport of Pi, ADP and ATP across the inner mitochondrial membrane), the **ATP balance is approximately 2.5 ATP/NADH and 1.5 ATP/$FADH_2$**

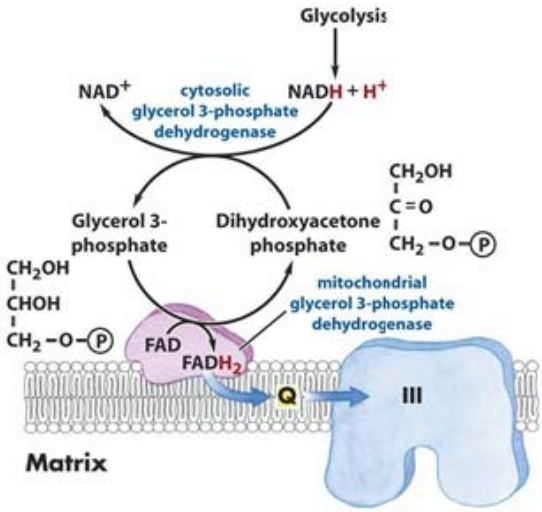
Shuttle systems to convey cytosolic NADH into mitochondria for oxidation

- At this time it is necessary to remember that **aerobic glycolysis is highly dependent on the availability of NAD^+** (glyceraldehyde 3-phosphate dehydrogenase reaction).



- The regeneration of NAD^+ **requires that NADH be reoxidized in the mitochondrial matrix by complex I** of the electronic transport chain
- Since the **inner mitochondrial membrane is impermeable to NADH**, special shuttle systems are required to transport the reduction equivalents from cytosolic NADH to the mitochondria
- **The most active shuttle, which functions in liver, kidney and heart mitochondria, is the malate-aspartate shuttle**
- This metabolic cycle introduces the electrons of NADH into the mitochondria **in the form of malate**. Subsequently, through the oxidation of malate to oxaloacetate, **electrons reduce mitochondrial NAD^+ to NADH**
- Those electrons will eventually be transferred to the electron transport chain and will contribute to the synthesis of 2.5 ATP

- **In skeletal muscle and brain they use the glycerol 3-phosphate shuttle**
- This shuttle differs from the previous one in that it yields the reduction equivalents from NADH to ubiquinone, so it does not use the entry gate of complex I and avoids one coupling center, which leads into 1 ATP less (1, 5 ATP)



Regulation of the oxidative phosphorylation

- Since oxidative phosphorylation is responsible for most of the ATP produced in cells, it is essential that ATP synthesis is regulated to adapt to variable ATP needs
- **In general terms, oxidative phosphorylation is regulated by the availability of ADP**.

- When the consumption of ATP is triggered in the cell as a consequence of the activation of anabolic processes, the increase in [ADP] triggers the consumption of O_2 and therefore the synthesis of ATP

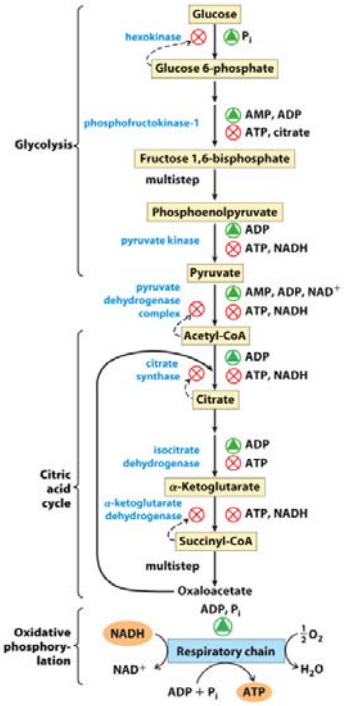


Figure 19-35
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Protective mechanism against the ATPase activity of ATP synthase under hypoxia

- At certain circumstances, ATP synthase can behave as a proton pump driven by the hydrolysis of ATP
- When a cell is hypoxic, such as during a heart attack or a cerebral stroke, the transfer of electrons to oxygen ceases and also the pumping of protons, which means that the proton gradient disappears.
- Under these conditions ATP synthase could operate as ATPase causing massive ATP depletion.
- This phenomenon is avoided by the dimerization (activation) of a protein inhibitor **called IF1**.
- The drop in pH that occurs during hypoxia as a consequence of the production of pyruvate and lactate (anaerobic glycolysis, favors the dimerization of IF1 in the mitochondrial matrix, which leads to the inhibition of the hydrolytic activity of ATP synthase, thus avoiding the hydrolysis of ATP

Defense mechanisms against ROS production during hypoxia

- During hypoxia there is an imbalance between the entry of electrons to the respiratory chain and the transfer of electrons to O_2 (absent in these conditions)
- This leads to an increase in the formation of ROS
- To avoid this, in addition to the glutathione peroxidase system, cells have two aditional defense mechanisms that lead to an **inhibition of PDH and the replacement of complex IV by another more effective in this situation**.
- Both effects are caused by and increased transcription of the hypoxia-inducible factor (HIF1), a transcription factor that activates the transcription of other enzymes and proteins..

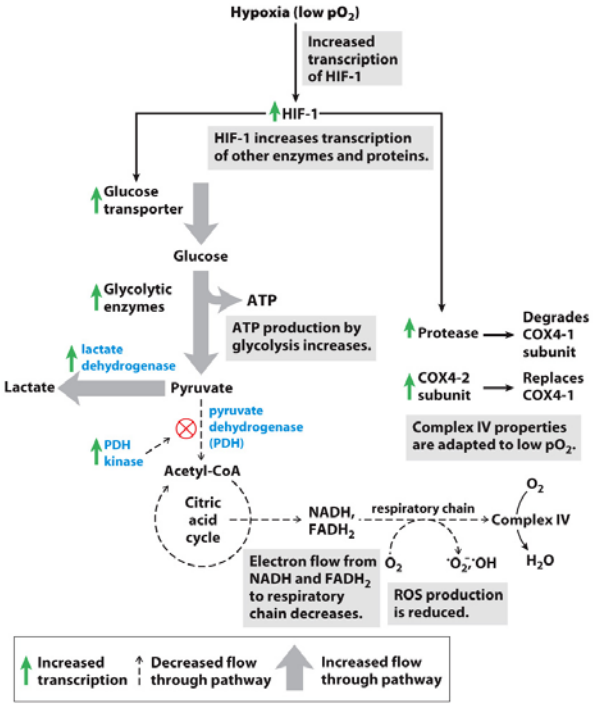


Figure 19-34
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